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CPRE 4880 - Embedded Systems Design

Course Project Proposal

For this semester's course project we want to design and implement a multi-channel audio looping station on an FPGA. This system will be able to record, store, playback multiple audio loops simultaneously, and ultimately function similarly to these widely used examples: [example 1](#) [example 2](#).

Requirements

- Record audio input and store it as a looping clip on any of 4 independent channels.
- Play back all active channels simultaneously with seamless, gapless looping.
- Support overdub: layer new audio on top of an already-playing loop.
- Accept audio from 2 independent input sources simultaneously.
- Provide per-channel volume control adjustable by the user in real time.
- All channel state changes (record / play / stop / overdub) triggered via onboard buttons.
- Audio sample rate of 48 kHz, 16-bit on all channels.
- Minimum 1 minute of loop storage per channel using onboard DDR3 RAM

Potential Challenges

- Clock domain crossing - the I2S codec clocks and FPGA fabric clock are asynchronous. Proper CDC FIFOs are required to avoid metastability and sample corruption between domains.
- Loop synchronization - the record end point must snap to a clean sample boundary, otherwise a click or pop will be audible at every loop repeat.
- AXI DMA integration - routing recorded samples from the PL capture buffer through the PS into DDR3 via DMA adds design complexity and requires careful management of buffer addresses and transfer sizes.
- DDR3 latency Multi-source clock alignment - driving both the onboard codec and PMOD I2S2 from the same MMCM-generated master clock is essential to prevent inter-source pitch drift over time.
- Button debounce and latency - mechanical buttons require debounce, and perceived latency between pressing record and audio capture must be minimized.
- Real-time mixing/passing -

Expected Resources

Item	Use
Zeboard Zynq-7000Q	Computation device
PMOD I2S2	Additional stereo audio input source
1/4" to 3.5mm passive adapter	To adapt to a wide range of input sources

Onboard pushbuttons and LEDs	User interface and status feedback
Audio cables and headphones / speaker	Output monitoring

Grading Rubric			
Attributes	1: Beginning - Unsatisfactory - Low level	2: Accomplished - Satisfactory - Medium Level	3: Exemplary - Beyond Satisfactory - High level
Audio Playback	Audio loops continuously and play back for one-channel [20 points]	Audio continuously loops and playbacks multiple-channels of audio [40 points]	Each audio channel has independent volume control [60 points]
Audio Recording	Audio can record one-channel [20 points]	Audio can be recorded onto multiple-channels (stacked) through multiple sources [40 points]	Audio can be recorded onto multiple-channels from multiple sources [60 points]
User operation and playability	Limited usefulness, is clunky and difficult to use [20 points]	Acceptable usefulness, user can easily record / playback and loop audio [40 points]	User can seamlessly record / loop from multiple audio sources, control channel playback gain (volume) and overdub [60 points]
Memory and Performance	Monitor Latency is very apparent > 20 ms. Audio storage capacity is limited < 10 seconds [20 points]	Latency is acceptable > 15ms < 10 ms. Can record at least 1 minute of audio per channel (4 minutes total). [40 points]	Latency is negligible and barely noticeable > 10 ms. Can record at least 2 minutes of audio per channel (8 minutes total) [60 points]
Demo and report	Limited demo and report [20 points]	Full demo, report includes description of all major components [40 points]	Entertaining demo, report also includes detailed figures and system evaluation results [60 points]